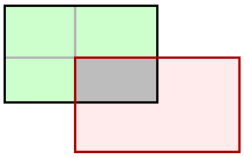
Lab 1 – math questions

Instructions

- Choose one of the questions below.
- Read the problem and examples from the given link.
- Solve the problem
- Submit the solution or partial solution to Moodle.
- Make sure that your submission includes:
    - Question name
    - Link to the question website
    - Problem description (Copy&paste the question site is ok)
    - Explanation and thoughts about the solution including time and space complexity
    - The solution



1. **Total Moves For Bishop**
   https://www.interviewbit.com/problems/total-moves-for-bishop/
   Given the position of a Bishop (A, B) on an 8 * 8 chessboard, count the total number of squares that can be visited by the Bishop in one move. The position of the Bishop is denoted using row and column number of the chessboard.
   Note: Bishop (רץ) can travel to any distance diagonally.



2. **Rectangle Area**
   https://leetcode.com/problems/rectangle-area/
   Given the coordinates of two rectangles in a 2D plane, return the total area covered by the two rectangles.
   First rectangle: bottom-left corner (ax1, ay1), top-right corner (ax2, ay2).
   Second rectangle: bottom-left corner (bx1, by1), top-right corner (bx2, by2).

3. **Permutation Sequence**
   https://leetcode.com/problems/permutation-sequence/
   The set [1, 2, 3, ..., n] has a total of n! unique permutations. By listing and labeling all of the permutations in order, we get the following sequence for n = 3: "123", "132", "213", "231", "312", "321".
   Given n and k, return the kth permutation sequence.

4. **Reverse Integer**
   https://leetcode.com/problems/reverse-integer/
   Given a signed 32-bit integer x, return x with its decimal digits reversed.
   Note: If reversing x causes the value to go outside the signed 32-bit integer range $[-2^{31}, 2^{31} - 1]$, then return 0.

5. **Power of 2**
   https://www.interviewbit.com/problems/power-of-2/
   Given a number in decimal string format, find if it is a power of 2 or not.
   Note: The input number may be too large to fit in 64 bits.